Sam Herrera | Aerospace & Mechanical Engineer

samueljherrera@att.net | (618) 567-6280 | linkedin.com/in/sam-herrera-ba41652b3 | sam-herrera111.github.io/portfolio

EDUCATION

Washington University in St. Louis St. Louis, MO
Master of Science in Aerospace Engineering **Expected Dec 2027**
Bachelor of Science in Mechanical Engineering **Expected May 2027**
Cumulative GPA: 3.86 *Dean's List: Spring 2024 – Present*
Coursework: Aerodynamics (enrolled), Finite Element Analysis (enrolled), Flight Dynamics, Aircraft Design, Composites (enrolled), Aircraft Propulsion (enrolled)

WORK EXPERIENCE

Mechanical Engineering Intern — Salas O'Brien **Summer 2026**
- Calculated thermo-fluid heating/cooling loads, pressure drop (ΔP) dynamics, and hydronic pump sizing across 3 facility infrastructure projects valued over $2M+.
- Resolved multi-discipline mechanical, structural, and plumbing layout interferences, cutting design revision turnaround by 20% while ensuring 100% code compliance.

Teaching Assistant — Fluid Mechanics — Washington University in St. Louis **Fall 2026 – Present**
- Instruct 70+ engineering students on Navier-Stokes equations, differential boundary layer theory, laminar-to-turbulent transition, and similitude.
- Lead weekly technical recitations and laboratory sessions covering manometer hydrostatics, flow visualization, and aerodynamic drag verification.

Process Mechanical Engineering Intern — Procter & Gamble, Cincinnati, OH **Summer 2025**
- Modeled tooling retrofits in CAD and performed structural FEA to mitigate dynamic fatigue, reducing unplanned line downtime by 12%.
- Executed mechanical upgrades on high-speed automated lines, cutting tooling changeover setup time and boosting throughput by 2%.

ENGINEERING LEADERSHIP & PROJECTS

Chief Engineer — WashU Design/Build/Fly (WUDBF) **Summer 2026 – Present**
- Direct aircraft architecture, flight envelope definitions, and mass margins ($<$15 lbs MTOW target) across aero, structures, propulsion, and avionics.
- Enforce structural margins of safety (MS > 0) using non-linear FEA and standardize multi-layer carbon fiber composite vacuum bagging layups.
- Coordinate flight testing programs to evaluate static/dynamic stability derivatives and onboard telemetry data across 80+ team members.

Aerodynamics & Payload Sub-Team Member/Lead — WashU Design/Build/Fly **Fall 2023 – Spring 2026**
- Executed 2D/3D CFD sweeps (k-ω SST) in ANSYS Fluent and XFLR5 across $Re \approx 200$k–400k, raising maximum wing lift coefficient ($C_{L,\max}$) by 15%.
- Engineered high-lift Fowler flap kinematics and modular payload retention systems, securing 19th/107 (2024) and 35th/114 (2025) finishes globally.

Autonomous Personal Carrier **Spring 2026**
- Designed modular chassis in SolidWorks and fabricated 3D-printed load-bearing brackets optimized for stiffness-to-weight ratio.
- Integrated Arduino microcontroller with ultrasonic/IR sensor arrays and PID motor controllers for closed-loop autonomous path tracking.
- Validated physical prototype in under 4 weeks with 100% obstacle avoidance reliability, finishing 20% under allocated budget.

TECHNICAL SKILLS

CAD / FEA / CFD: SolidWorks, Siemens NX, AutoCAD, ANSYS (Fluent, Mechanical), XFLR5, COMSOL Multiphysics
Aerospace & Controls: Flight Dynamics, Airfoil Analysis, Stability Derivatives, MATLAB, Simulink, Arduino, dSPACE, GD&T
Programming & Fabrication: Python, C++, Java, Linux, LaTeX, CNC Milling, Lathe, Composites/Vacuum Bagging, 3D Printing
Languages: Fluent in Spanish (Bilingual)

INVOLVEMENT

American Institute of Aeronautics and Astronautics (AIAA) — *Member* **Fall 2023 – Present**
Society of Hispanic Professional Engineers (SHPE) — *Active Member* **Fall 2023 – Present**